

HW6

Part 1

We use pdfplumber to extract text and tables. Text is chunked with RecursiveCharacterTextSplitter (chunk_size=1000, overlap=150). Tables are preserved as single markdown chunks to avoid splitting rows/columns. Metadata added: page, type, table_index, heading (section title).

Part 2

Chunks are stored in a Chroma vector database using local Ollama embeddings. Retrieval supports metadata filters: tables-only, text-only, or all. Additional /tables endpoint enables keyword-based table browsing.

Part 3

Retriever.invoke fetches top-k chunks. Context is concatenated and passed to Ollama.

Prompt:

1. According to the 'Other income/(expense), net' table, what were interest and dividend income in 2020 and 2019?

Ask a Question (RAG)

Enter your question about the PDF:

g to the 'Other income/(expense), net' table, what were interest and dividend income in 2020 and 2019?

Restrict retrieval to:

All chunks

Ask

Answer

According to the 'Other income/(expense), net' table, interest and dividend income were:

- In 2020: \$3,763
- In 2019: \$4,961

Sources (from RAG)

Source 1 – page 49 (text)

Heading: (no heading detected)

Table index: None

Other Income/(Expense), Net The following table shows the detail of OI&E for 2020, 2019 and 2018 (in millions):

	2020	2019	2018
Interest and dividend income	3,763	4,961	5,686

Interest expense	(2,873)	(3,576)	(3,240)
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Other income/(expense), net	(87)	422	(441)
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Total other income/(expense), net	803	1,807	2,005
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Note 5 – Income Taxes U.S. Tax Cuts and Jobs Act On December 22, 2017, the U.S. enacted the Tax Cuts and Jobs Act (the “Act”), which significantly changed U.S. tax law. The Act lowered the Company’s U.S. statutory federal income tax rate from 35% to 21% effective January 1, 2018, while also imposing a deemed repatriation tax on previously deferred foreign income. The Act also created a new minimum tax on certain foreign earnings, for which the Company has elected to record certain deferred tax assets and liabilities. Provision for Income Taxes and Effective Tax Rate The provision for income taxes for 2020, 2019 and 2018,

2. In the ‘Other income/(expense), net’ table, was interest expense higher in 2020 or 2019?

Ask a Question (RAG)

Enter your question about the PDF:

In the "Other income/(expense), net" table, was interest expense higher in 2020 or 2019?

Restrict retrieval to:

All chunks

Ask

Answer

I don't know. The provided context does not mention "Other income/(expense), net" table or interest expense for specific years (2019 and 2020).

Sources (from RAG)

Source 1 - page 74 (table)

Heading: (no heading detected)

Table index: 2

of the 2042 Notes is July 31, 2042, and interest at a rate of 3.600% per annum is paid semi-annually on January 3

0 and July 31 of each year, beginning on January 31, 2016, and on the maturity date. As of

Explain Source 1

Source 2 - page 73 (table)

Heading: (no heading detected)

3. According to the Provision for Income Taxes table, what is Apple's effective tax rate in 2020?

According to the Provision for Income Taxes table, what is Apple's effective tax rate in 2020?

Restrict retrieval to:

All chunks

Ask

Answer

According to the context, Apple's effective tax rate in 2020 was 14.4%. This information can be found on page 46 of the 2020 Form 10-K report.

Sources (from RAG)

Source 1 - page 62 (text)

Heading: (no heading detected)

Table index: None

the account or disclosure to which it relates. Uncertain Tax Positions Description of the Matter As discussed in Note 5 to the financial statements, Apple Inc. is subject to taxation and files income tax returns in the U.S. federal jurisdiction and many state and foreign jurisdictions. As of September 26, 2020, the total amount of gross unrecognized tax benefits was 16.5 billion, of which 8.8 billion, if recognized, would impact Apple Inc.'s effective tax rate. Apple Inc. uses significant judgment in the calculation of tax liabilities in estimating the impact of uncertainties in the application of technical merits and complex tax laws. Auditing management's evaluation of whether an uncertain tax position is more likely than not to be sustained and the measurement of the benefit of various tax positions can be complex, involves significant judgment, and is based on interpretations of tax laws and legal rulings. Apple Inc. | 2020 Form 10-K | 59

Explain Source 1

4. Based on the 'Products and Services Performance' section, what were net sales by category for 2020, 2019, and 2018?

Ask a Question (RAG)

Enter your question about the PDF:

‘Products and Services Performance’ section, what were net sales by category for 2020, 2019, and 2018?

Restrict retrieval to:

All chunks

Ask

Answer

According to the context, the net sales by category for 2020, 2019, and 2018 are:

- iPhone (1): 137,781 (3%) in 2020, 142,381 (14%) in 2019, and \$164,888 in 2018
- Mac (1): 28,622 (11%) in 2020, 25,740 (2%) in 2019, and \$25,198 in 2018
- iPad (1): 23,724 (11%) in 2020, 21,280 (16%) in 2019, and \$18,380 in 2018
- Wearables, Home and Accessories (1)(2): 30,620 (25%) in 2020, 24,482 (41%) in 2019, and \$17,381 in 2018
- Services (3): 53,768 (16%) in 2020, 46,291 (16%) in 2019, and \$39,748 in 2018
- Total net sales: 274,515 (6%) in 2020, 260,174 (-2%) in 2019, and \$265,595 in 2018

Sources (from RAG)

> Source 1 – page 24 (text)

> Source 2 – page 24 (text)

Search Tables

Part 4 - Experiments & Evaluation

I evaluated my RAG system using several questions drawn from Apple’s 2020 Form 10-K (uploaded as NASDAQ_AAPL_2020.pdf).

My goal was to assess:

1. Extraction quality (text + tables)
2. Retrieval accuracy (Chroma + embeddings)
3. LLM reasoning accuracy (grounded answers vs hallucinations)
4. behaviors (metadata filtering, table-only mode, explanations)

Below, I summarize success, partial success, and failure cases.

1. “‘Products and Services Performance’ section, what were net sales by category for 2020, 2019, and 2018?”

Model Answer:

iPhone: 137,781 (2020), 142,381 (2019), 164,888 (2018)

Mac: 28,622 ; 25,740 ; 25,198

iPad: 23,724 ; 21,280 ; 18,380

Wearables: 30,620 ; 24,482 ; 17,381

Services: 53,768 ; 46,291 ; 39,748

Total net sales: 274,515 ; 260,174 ; 265,595

Ground Truth (Page 24 of PDF)

NASDAQ_AAPL_2020

Matches exactly the values in Apple's official "Net sales by category" table.

Evaluation:

This is a high-value success case:

The system successfully retrieved the correct narrative + table chunk.

It correctly interpreted a five-row, multi-year financial table.

Numeric extraction was entirely accurate.

Retrieval was effective even without "tables-only" mode.

This demonstrates that the ingestion pipeline, chunking strategy, and reranking logic all work for structured financial tables.

2. "According to the 'Other income/(expense), net' table, what were interest and dividend income in 2020 and 2019?"

Model Answer:

2020: \$3,763

2019: \$4,961

Ground Truth (Page 27 of PDF)

NASDAQ_AAPL_2020

Correct values appear in the OI&E table.

Evaluation:

Retrieval surfaced the correct OI&E chunk.

The LLM correctly extracted numbers.

This verifies that table preservation + markdown formatting allows the model to interpret numeric rows reliably.

3. "According to the Provision for Income Taxes table, what is Apple's effective tax rate in 2020?"

Model Answer:

“Apple’s effective tax rate in 2020 was 14.4%. This information is found on page 46.”

Ground Truth (Page 27–28 of PDF)

NASDAQ_AAPL_2020

Correct: The effective tax rate is indeed 14.4%.

Evaluation:

The model retrieved an adjacent narrative chunk (not the table itself).

Even without the exact table, the model extracted the correct number.

This is a partial success: correct answer, but retrieval could be improved to grab the actual table block.

4. “In the ‘Other income/(expense), net’ table, was interest expense higher in 2020 or 2019?”

Model Answer:

“I don’t know. The provided context does not mention the OI&E table or interest expense for 2020 and 2019.”

Sources Retrieved:

Random debt-related tables (page 73–74).

Ground Truth (Page 27):

2020: \$(2,873)

2019: \$(3,576)

=> 2019 interest expense was higher.

Evaluation:

This is an example of embedding ambiguity:

The retriever pulled debt tables (also containing the word “interest”).

The correct OI&E table was not ranked highly enough.

Because the context lacked the correct numbers, the LLM responded with the safe answer “I don’t know”.

This is a good failure:

It shows retrieval weakness,

BUT it also shows the LLM remained grounded and avoided hallucination.

Part 5 - Streamlit UI

UI was built

Part 6 - Reflection

I'll be honest this project took way more out of me than I expected. When I first read the assignment, I thought, "Okay, extract the PDF, put it into a vector store, ask questions. Easy." But the moment I started actually working with a real 10-K document, everything I assumed went out the window.

I started off trying to do everything with Together AI (as it was paid) but I switched over to Ollama, and honestly, that changed everything. Being able to run the model locally, get instant responses, and actually see what's happening made the whole workflow so much smoother. I wish I had done it earlier.

The hardest part by far was dealing with the tables. Extracting text was easy. Tables was hard. Some tables would extract cleanly, others would come out scrambled or missing columns, and sometimes pdfplumber would just give me garbage rows. I had to rewrite my table extraction code multiple times until it finally hit me: just keep the tables whole and don't try to split them. Once I turned them into Markdown blocks, everything magically started making sense to the model. It was one of those moments where you're half relieved and half annoyed that you didn't think of it sooner.

Another big thing I learned is that the retriever matters more than the LLM. I kept thinking the model wasn't enough when really it was just being fed the wrong chunks. I saw it hallucinate less and say "I don't know" more once I tightened the context and the retrieval mode. The number of times the retriever grabbed the wrong "Net Sales" table made me realize how tricky embeddings can be when multiple tables look similar.

Building the Streamlit UI was actually fun compared to the rest. Adding the table browser, table-only mode, and the little "Explain Source" feature made everything way easier to debug. It also made the system feel interactive rather than this mysterious black box. Seeing the tables actually rendered properly in a DataFrame felt like a small victory after all the extraction drama.

Overall, this project taught me that RAG systems sound simple in theory but get messy in practice especially with financial documents. I learned a lot about chunking, metadata, embeddings, and how important it is to have good tooling around everything. It was frustrating at times, but by the end I actually felt proud of how much I figured out along the way.